



ASEAN Nexus
Case Study 8: AI for Inclusive MSME Growth

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Case Study 8: AI for Inclusive MSME Growth

1. Introduction

1.1 Background of Case Study

Micro, Small, and Medium Enterprises (MSMEs) are the backbone of ASEAN's economy. According to Case Study 8 in the BorNEO HackWknd 2026 booklet, MSMEs account for more than 97% of all business establishments and provide approximately 85% of employment in the region. However, despite their immense economic importance, MSMEs contribute only around 45% of ASEAN's GDP and 18% of export value, revealing a significant productivity and digital participation gap across the region.

This disparity largely stems from systemic barriers to formalization, digital adoption, and access to capital. Many micro-entrepreneurs operate with fragmented business knowledge, lack structured operational planning, and face difficulties navigating both private and government financial ecosystems. Without centralized and actionable guidance, these businesses struggle to scale sustainably and remain competitive in increasingly digital markets.

To address this challenge, this project introduces an AI-powered consultative platform designed to support the growth of ASEAN MSMEs. The application provides diagnostic business surveys, AI-generated growth roadmaps, financial tracking tools, and an integrated Retrieval-Augmented Generation (RAG) assistant to guide entrepreneurs through practical business decisions. By simplifying access to business intelligence and digital tools, the platform helps informal micro-businesses **transition into structured, competitive, and revenue-generating enterprises.**

1.2 Sustainable Development Goals Implemented

1.2.1 SDG 8: Decent Work and Economic Growth

This application supports SDG 8 by empowering Micro, Small, and Medium Enterprises (MSMEs), which form the backbone of the ASEAN economy. Many micro-business owners face challenges such as limited business knowledge, lack of structured planning, and difficulty accessing financial support. The proposed application addresses these issues by providing personalized growth roadmaps with step-by-step guidance to help entrepreneurs improve operations, marketing, and financial management.

Additionally, the system enhances financial accessibility by connecting users with relevant government and private loan opportunities while offering built-in financial tracking to monitor revenue and expenses. By enabling MSMEs to make informed business decisions and scale sustainably, the application promotes **entrepreneurial empowerment, economic participation, and long-term business growth**.

1.2.2 SDG 9: Industry, Innovation, and Infrastructure

This application promotes the technological upgrading and digital transformation of traditional micro-businesses, giving them the tools to compete in modern markets. Many MSMEs in the ASEAN region still rely on manual, fragmented processes that severely limit their scalability. This platform acts as a crucial digital bridge by providing an accessible, centralized infrastructure for business management. Through the customized roadmaps, the application actively recommends specific, modern software tools, accelerating technology adoption and transitioning these enterprises toward efficient, digital-first workflows.

Furthermore, the integration of a Retrieval-Augmented Generation (RAG) AI bot democratizes access to high-level business intelligence and innovation. This AI assistant provides on-demand, contextual troubleshooting and tailored guidance, serving as an advanced technological infrastructure that was previously out of reach for micro-entrepreneurs. By equipping MSMEs with data-driven insights and scalable technological frameworks, the application fosters a culture of **continuous innovation, digital resilience, and enhanced market competitiveness**.

1.3 Problem Statement

MSMEs form the economic foundation of ASEAN, they face critical structural barriers:

1. Fragmented and Generic Business Guidance

Existing resources provide high-level, generic information, leaving micro-entrepreneurs without actionable, step-by-step execution plans tailored to their specific industry and growth phase.

2. Challenge Faces of Capital Access

Navigating the complex landscape of private and government funding is difficult, and many micro-businesses lack the structured financial data required to secure vital growth capital.

3. Lack of Digital Integration

Traditional MSMEs often rely on manual, disconnected processes and lack the technical knowledge to select and implement modern digital tools required for operational efficiency.

4. Real-Time Contextual Help Support Issue

When facing immediate operational or regulatory roadblocks, micro-businesses cannot afford consultants and lack a reliable, localized source for troubleshooting and advice.

5. Poor Visibility in Financial Tracking and Goal Motivation

Without accessible tools to track current revenue against target growth goals, MSMEs struggle to maintain the long-term sustainability, financial discipline, and motivation necessary for expansion.

1.4 Objectives

The objectives of our project are:

1. Provide Actionable Business Intelligence

- Translating generic advice into tailored, step-by-step execution plans.

2. Accelerate Digital Transformation

- Equipping traditional micro-businesses with modern, accessible tech tools.

3. Deliver On-Demand Contextual Support

- Providing 24/7 localized troubleshooting via our AI Business Coach.

4. Drive Sustainable Financial Discipline & Capital Access

- Helping owners set, track, and hit revenue targets.

5. Facilitate Access to Growth Capital & Resources

- Connecting investment-ready MSMEs with grants and formal financing.

2. Review of Existing Solutions

2.1 Comparison in between Industry Landscape

Software Categories	ASEAN Gov. Portals	Global Business Planning	SME Accounting & ERP	Our Application (ASEAN Nexus)
Primary Purpose	Information, Policies & Grant Directories	Strategic Business Plan Generation	Historical Financial Tracking & Invoicing	End-to-End Execution & Capital Matching
Step-by-Step Execution Roadmaps	No (Static articles only)	High-level only (focuses on writing a document)	No	Yes (Gamified, actionable micro-tasks)
ASEAN-Specific Grant Matching	Yes, but user must search manually	No	No	Yes (Auto-matched based on current milestone)
On-Demand AI Co-Pilot (RAG)	Basic FAQ chatbots	Yes (But lacks local ASEAN context)	No	Yes (Trained on local compliance & grants)
Financial Scenario Planning	No	Yes (only financial tracking purpose)	Basic forecasting	Yes (With live Bank-Readiness Scoring)
Contextual Tool Recommendations	Broad, uncurated lists	No	No	Yes (Curated at the exact step needed)

2.2 Comparison in between competitors with specifics Feature/System

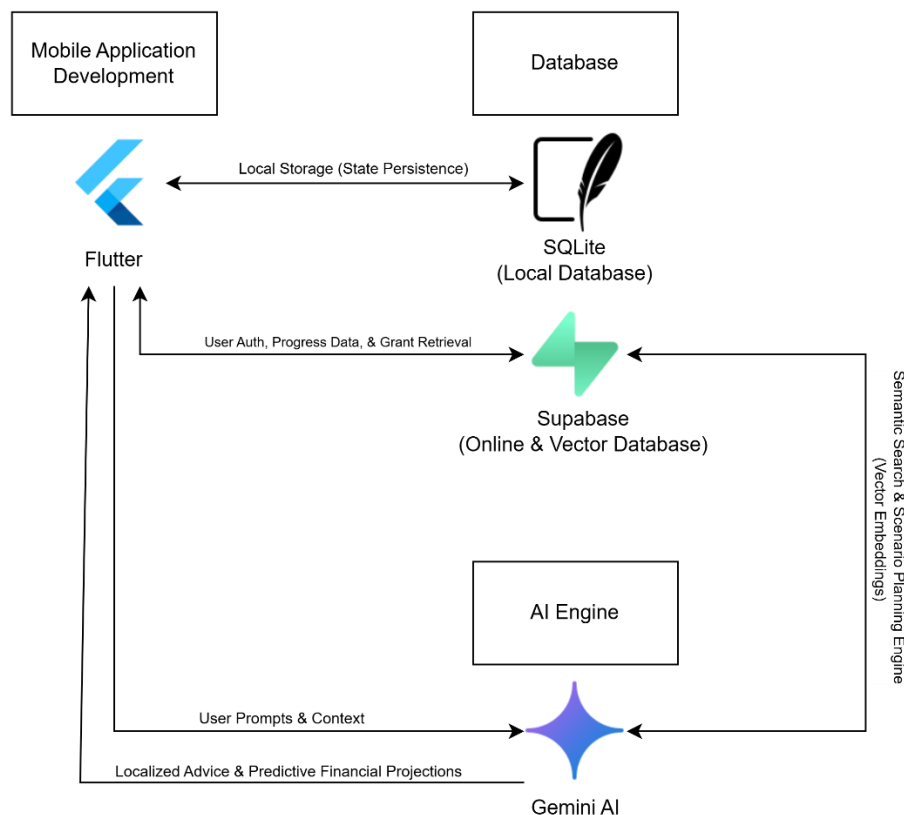
System	LivePlan	Upmetrics	IdeaBuddy	Our Application (ASEAN Nexus)
Target Users	Startups & SMEs	Entrepreneurs	Early-stage founders	ASEAN MSMEs
AI Guidance	Limited	AI plan generation	Idea validation	AI-generated growth roadmap
Financial Tracking	Yes	Yes	Basic	Integrated financial tracking
Loan / Funding Access	No	No	No	Government & private loan access
Regional Focus	Global	Global	Global	ASEAN-focused support
AI Assistant	No	Limited	No	RAG-based business advisor
Development	Website	Website	Website	Android & IOS (Flutter)

3. Methodology

3.1 System Architecture

The ASEAN Nexus platform operates on a modernized, serverless architecture. The client-side Mobile Application is built with Flutter, ensuring cross-platform compatibility. It uses SQLite for local state persistence, allowing for offline data caching.

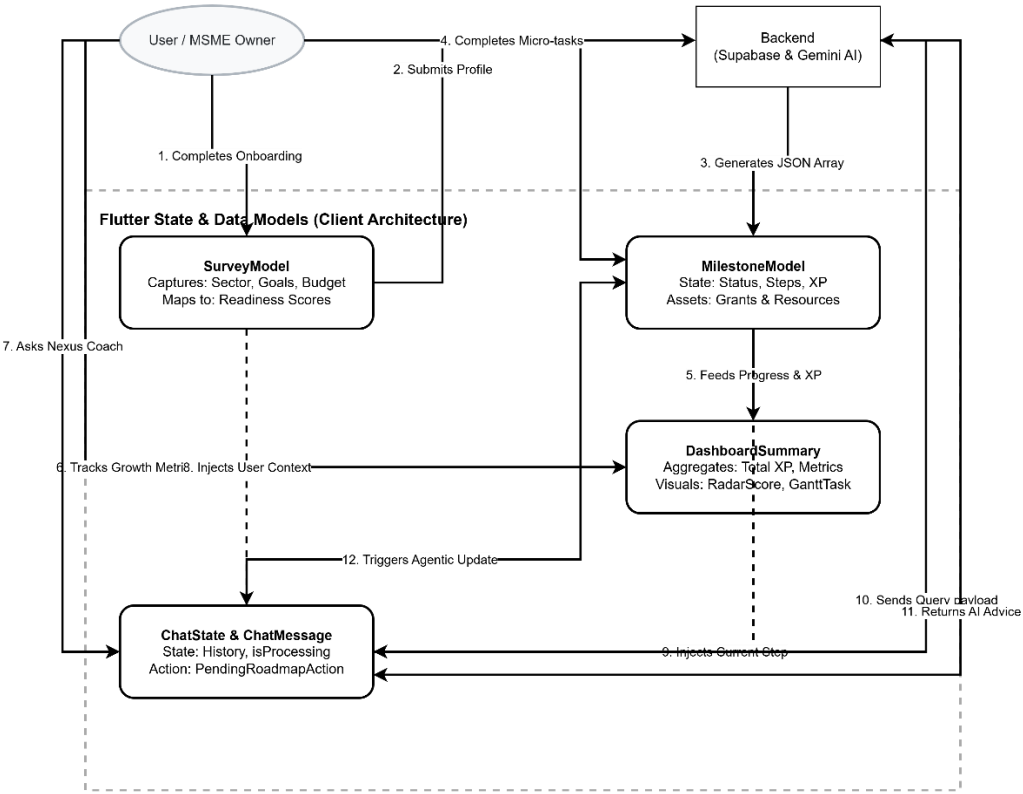
The primary backend is powered by Supabase, which handles secure user authentication, stores structured progress data, and acts as a vector database for matching relevant business grants. Finally, the Gemini AI Engine serves as the core intelligence layer. It receives user prompts and context from the app and performs semantic searches against the Supabase vector embeddings to return localized advice, predictive projections, and actionable roadmap steps.



3.2 Data Flow Diagram (Ai Prompting Flow)

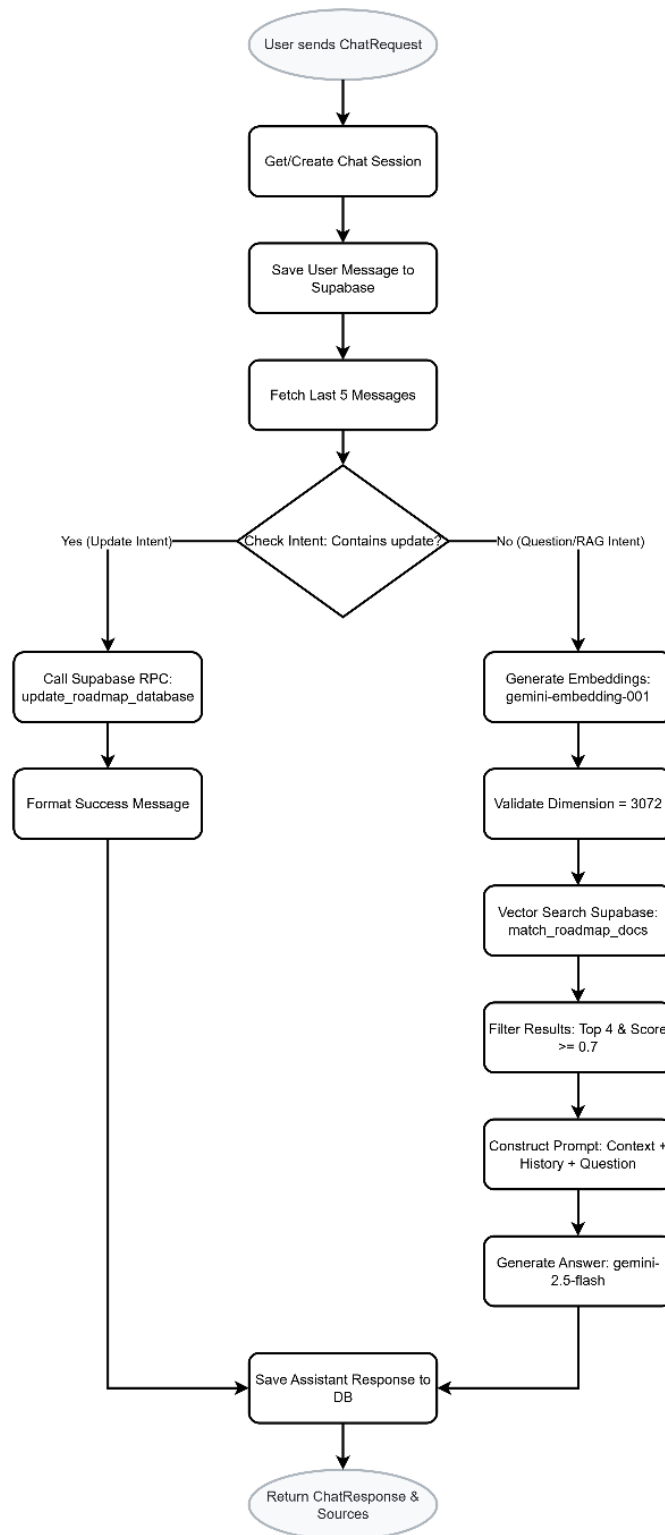
The system orchestrates data between the user, the client-side state, and the AI backend in a twelve-step process to generate personalized guidance.

System Stage	Explanation
Onboarding & Profiling	The User completes onboarding, submitting their business profile which is captured by the SurveyModel. This data informs the user's "Readiness Scores"
Roadmap Generation	The backend (Supabase & Gemini AI) processes the profile and generates a JSON array, populating the MilestoneModel with actionable steps, grants, and resources.
Execution & Tracking	As the User completes micro-tasks, progress is fed into the Dashboard Summary, updating growth metrics and visual indicators like the Radar Score.
Contextual Support	When the User asks the "Nexus Coach" for help, the app injects their current user context (from the Survey Model) and current step (from the Dashboard Summary) into the ChatState & ChatMessage model.
AI Query & Response	This combined payload is sent as a query to the backend. The Gemini AI processes the context and returns tailored advice.



Data Flow Diagram (ASEAN Nexus)

3.3 RAG Process Flowchart



4. Results

4.1 Evidence of Objective Achievement

4.1.1 ASEAN Nexus Use Case Diagram

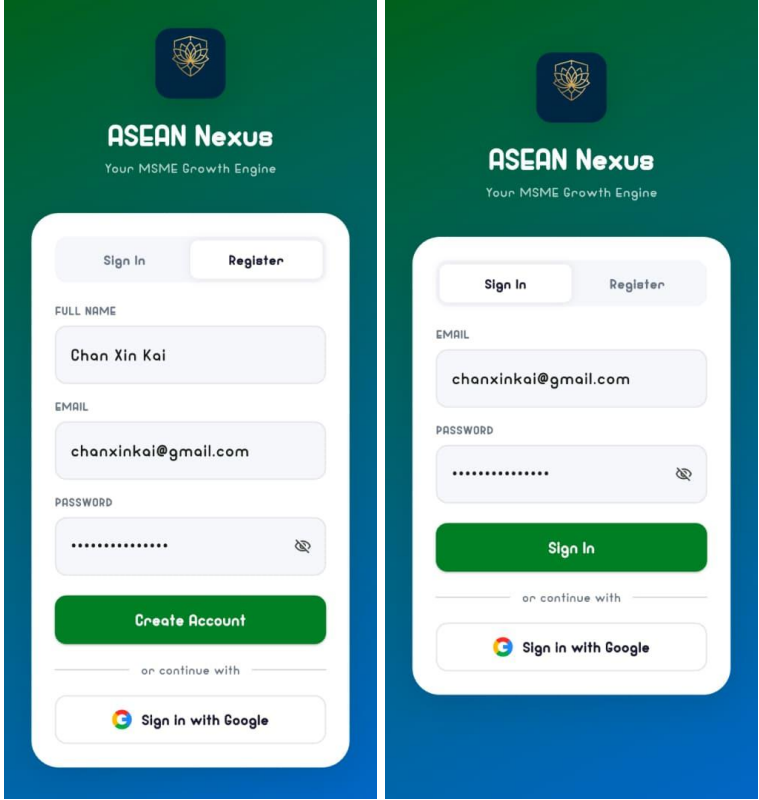
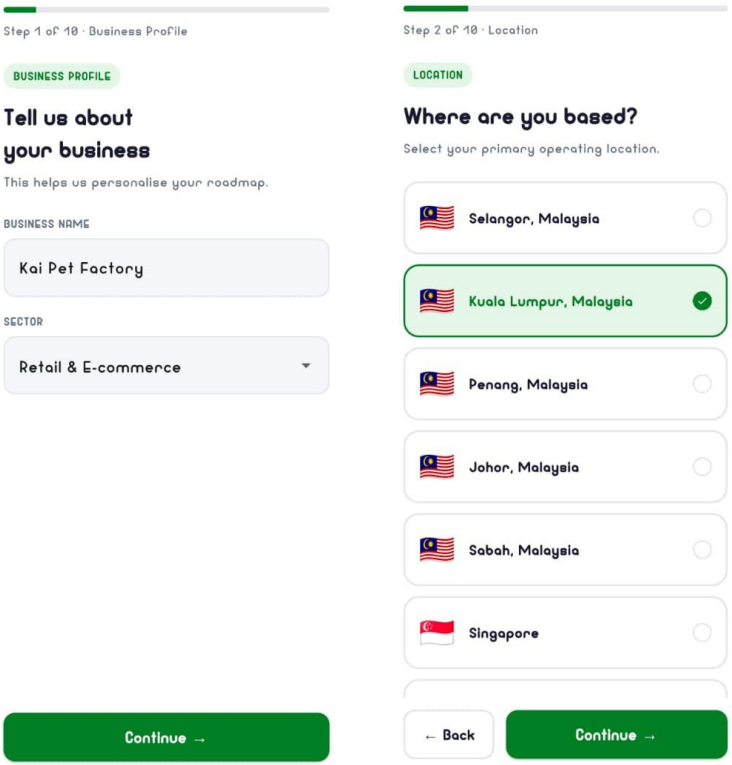


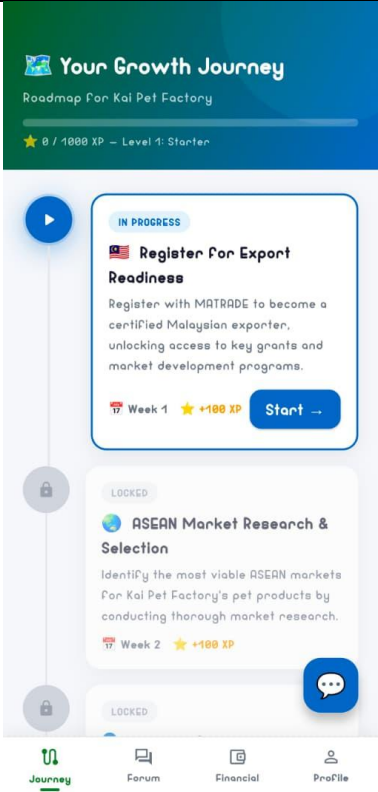
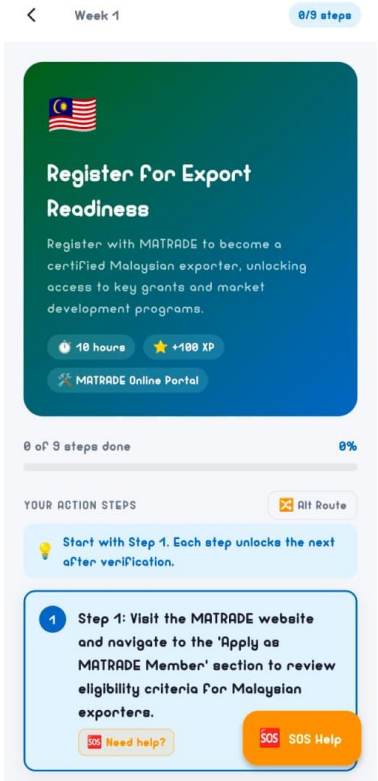
4.1.2 Use Case Objective Fulfilment

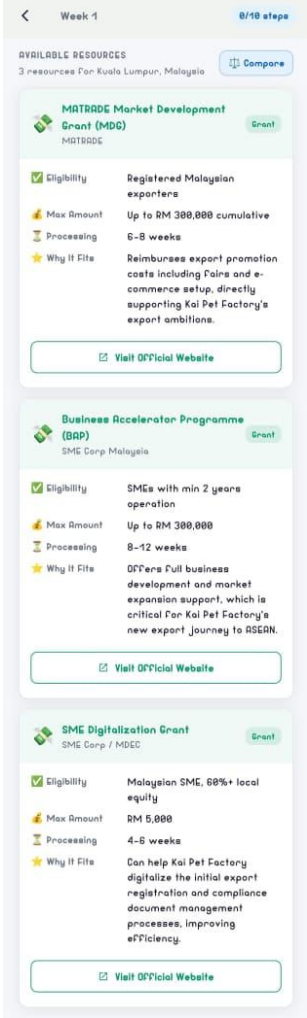
Objective Of Project	Use Case Implement
Provide Actionable Business Intelligence	• Answer Business Survey • Generate Dynamic Roadmap
Accelerate Digital Transformation	• Execute steps with Tool Suggestions
Deliver On-Demand Contextual Support	• Consult AI Business Coach
Drive Sustainable Financial Discipline & Capital Access	• Set & Monitor Financial Targets
Facilitate Access to Growth Capital & Resources	• Analyze Resources Options • Track Milestone Progress

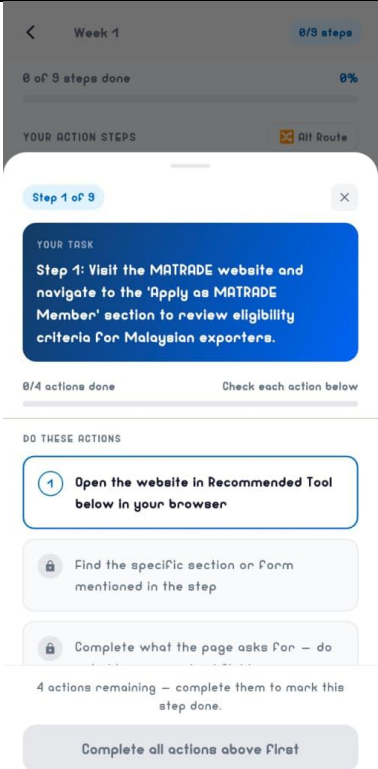
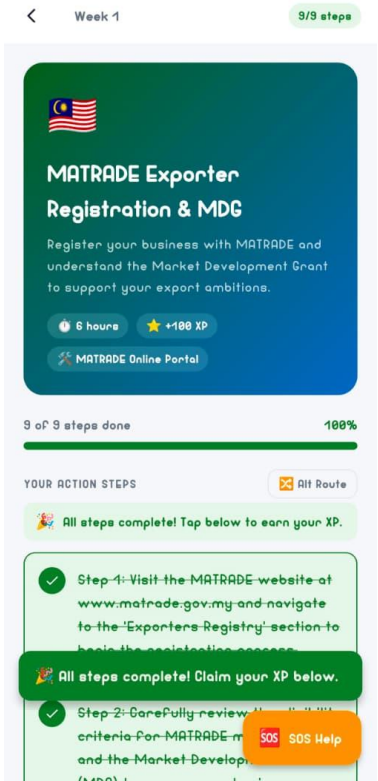
The following table demonstrates how the core objectives of the project are directly fulfilled by the implemented use cases within the application's architecture.

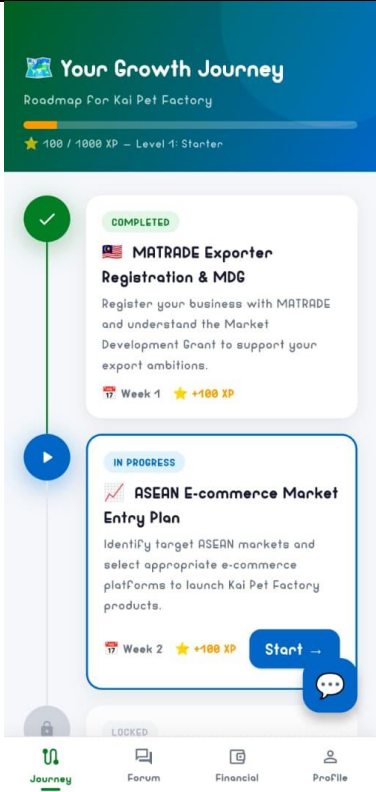
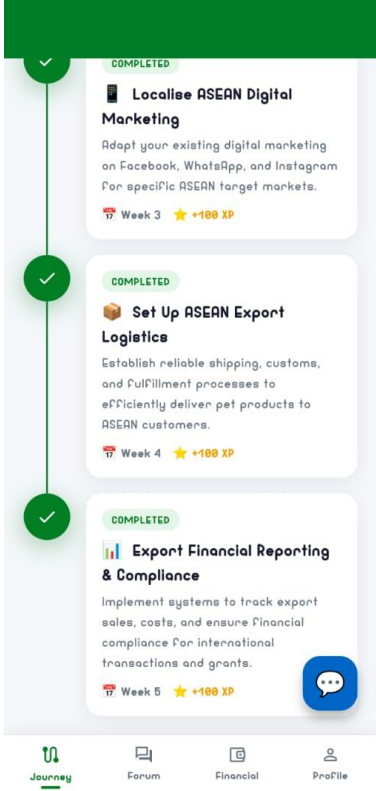
5. Prototype Showcase

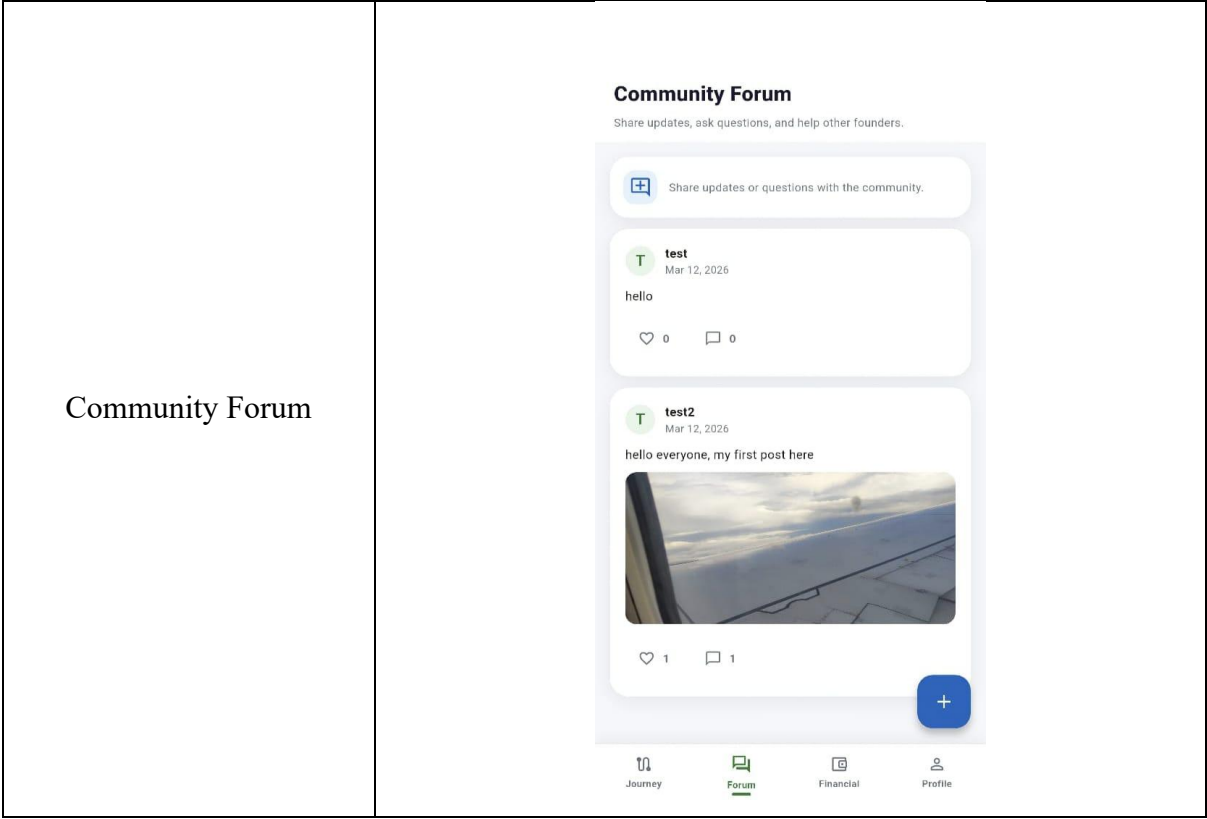
Sign in / Register New Account	
Survey Questions (Total 10 Questions)	

Growth Journey	
Week 1 Task	

Loan Type Comparison	
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Action Steps	
Earn Points	

Continue Week 2	
All Tasks Completed	



Financial Co-Pilot

Financial Co-Pilot

Track monthly targets, daily actuals, and growth trend.

Mar 2026

Monthly Plan

Set your monthly budget and target growth.

Monthly Budget (RM)

2000

Target Growth (%)

2.0

Save Monthly Target

Projection vs Actual Growth

Budget RM 2000 | Target 2.0%



Projection Growth

Actual Growth

Daily Revenue & Expense

Date: 2026-03-13

Revenue (RM)

Expense (RM)

Note (optional)

Save Daily Log

Recent Daily Entries

2026-03-16

Rev RM 3000 | Exp RM 600

2026-03-13

Rev RM 100 | Exp RM 20

2026-03-12

Rev RM 600 | Exp RM 120

Grant & Funding Matcher

Matched by sector, location, and readiness.

GRANT

micro

RM 80000

Agrobank Agropreneur Grant 2026

Agrobank

2026-08-31

View

GRANT

Agropr

Ministry

2026-

Bank-Readiness Score

Need 47 more points to reach 80.

33%

Early Stage

Missing Requirements

Business registration

Register your business (UBIN/SSM) to unlock lender trust.

Score impact: missing 25.0 / 25.0 points (current 0.0).

Digital statement history

Need 6 more months of digital statements to maximize scoring.

Score impact: missing 20.0 / 20.0 points (current 0.0).

Digital sales channels

Add Shopee/Grab/TikTok Shop channels to strengthen digital proof.

Score impact: missing 10.0 / 20.0 points.

Journey

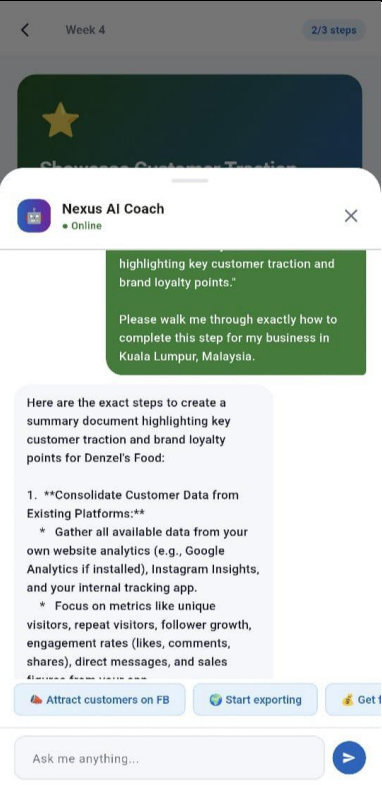
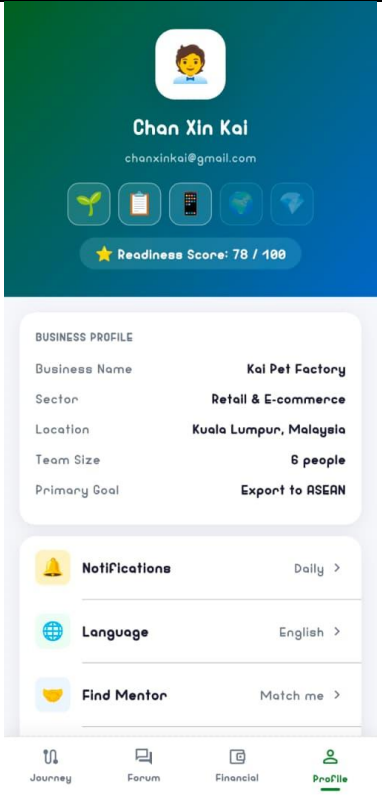
Forum

Financial

Profile

Team Journey2DeWest

UMPSA | March 2026

AI Bot	
Profile Page	

6. AI Acknowledgement

All core architecture, coding implementation, UI design, and validation were completed by the team.

AI Tools Used	Usage/Description
ChatGPT	● Project research regarding MSME topics, pain points, competitions, brainstorming ideas, and structuring concepts ● Debugging code and troubleshooting technical issues during development. ● Helped generate foundational code templates to accelerate development phases.
GEMINI	● Project research regarding MSME topics, pain points, competitions, brainstorming ideas, and structuring concepts ● Assisted in refining roadmap logic explanation. ● Assisted in structuring code snippets and optimizing development workflows

7. Conclusion

The Journey2DeWest team's report for the BorNEO HackWknd 2026 presents "ASEAN Nexus," an AI-powered consultative platform designed to support the growth of Micro, Small, and Medium Enterprises (MSMEs) in the ASEAN region. While MSMEs form the backbone of the ASEAN economy, they frequently struggle with fragmented business guidance, limited access to capital, and poor digital integration. To overcome these structural barriers, the ASEAN Nexus application provides users with diagnostic business surveys, personalized AI-generated growth roadmaps, financial tracking tools, and an integrated Retrieval-Augmented Generation (RAG) AI business coach for on-demand troubleshooting.

Developed through a structured, three-phase methodology using a Flutter mobile frontend, Supabase for database management, and the Gemini AI engine, the platform successfully meets its core objectives: providing actionable business intelligence, accelerating digital transformation, delivering contextual support, driving financial discipline, and facilitating access to growth resources. The final deployable prototype demonstrates a complete user journey, from initial onboarding to milestone tracking and AI consultation. The project concludes that the ASEAN Nexus solution effectively bridges the digital and productivity gaps for micro-entrepreneurs, for future to enhance the AI's knowledge base, integrate predictive financial forecasting tools, and implement a mentor matchmaking feature to further empower the regional MSME ecosystem.

Through phased development, the plan to build:

Feature for improvement	Explanation
Enhance RAG Knowledge Base:	Continuously expand the Retrieval-Augmented Generation (RAG) database with localized, industry-specific data to provide even more precise and actionable business intelligence.
Refine AI Prompt Engineering	- Optimize system prompts to improve the nuance, accuracy, and contextual relevance of the AI Business Coach's responses. - Improve the speed for the roadmap generated.
Optimize UI/UX for Resource Portals	Redesign the interface for analyzing resource options (grants, loans, etc.) to ensure a smoother, more intuitive user journey when comparing and selecting funding.
Implement Advanced Financial Tools	Predictive Revenue Analytics: Introduce tools that analyze historical data to forecast future sales trends. AI-Based Financial Forecasting: Integrate intelligent modelling to help MSMEs anticipate cash flow needs and optimize budget allocation.
Develop a Mentor Matchmaking Feature	Create a system to connect MSME owners with experienced industry mentors for personalized, human-in-the-loop guidance.